

Auteur: Viki Lauvrys
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$$z^4 = -1 = 1 \cdot \text{cis}(\pi)$$

$$\Rightarrow z_k = \text{cis}\left(\frac{\pi + 2k\pi}{4}\right) \text{ met } k \in \mathbb{Z}$$

De vierdemachtswortels zijn dan:

$$\Rightarrow z_0 = \text{cis}\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}i}{2}$$

$$\Rightarrow z_1 = \text{cis}\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i}{2}$$

$$\Rightarrow z_2 = \text{cis}\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}i}{2}$$

$$\Rightarrow z_3 = \text{cis}\left(\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}i}{2}$$

References